

Dynamic contour allocation for evidence and posterior resolution

A compact first-principles note

1. Setup and the two downstream kernels

Let the current classic tree define likelihood contours

$$L_0 < L_1 < \dots < L_G,$$

with corresponding prior-mass locations

$$1 = X_0 > X_1 > \dots > X_G.$$

Shell j is the interval $(X_j, X_{j-1}]$, with width

$$\Delta X_j := X_{j-1} - X_j, \quad z_j := L_j \Delta X_j, \quad m_j := \frac{z_j}{Z},$$

where $Z = \sum_j z_j$ and m_j is the posterior mass carried by shell j .

Suppose we may launch an extra constrained-sampling call inside any existing classic contour i . Under the ideal constrained prior, a draw from contour i is uniform in prior mass on $[0, X_i]$. This induces two natural kernels for all deeper shells $j > i$:

$$A_{ij} := \Pr(\text{draw from contour } i \text{ reaches boundary } j-1) = \frac{X_{j-1}}{X_i},$$

$$P_{ij} := \Pr(\text{draw from contour } i \text{ lands in shell } j) = \frac{\Delta X_j}{X_i}.$$

The kernel A_{ij} controls how much a draw from contour i can inform the shrinkage at deeper boundaries, while P_{ij} controls how much it contributes posterior support to shell j .

If one constrained call at contour i returns a whole correlated phantom cluster rather than a single point, let ρ_i denote its evidence-efficiency multiplier and let c_i denote its cost in likelihood evaluations. In the simplest case $\rho_i = 1$ and c_i is constant across contours.

2. Objective 1: reduce evidence uncertainty

Write the local shrinkage as $r_j = X_j/X_{j-1}$ and $t_j := \log r_j$. Since

$$Z = \sum_h L_h (X_{h-1} - X_h),$$

a perturbation in t_j changes the evidence by

$$\frac{\partial Z}{\partial t_j} = \sum_{h>j} L_h \Delta X_h - L_j X_j.$$

Define the dimensionless evidence sensitivity

$$g_j := \frac{1}{Z} \frac{\partial Z}{\partial t_j} = \frac{1}{Z} \left(\sum_{h>j} L_h \Delta X_h - L_j X_j \right).$$

If $v_j := \text{Var}(t_j)$, a delta-method proxy is

$$\text{Var}(\log Z) \approx \sum_j g_j^2 v_j.$$

Now let I_j be the current effective information count for boundary j , so that $v_j \propto I_j^{-1}$. A new draw from contour i contributes to boundary j only if it reaches boundary $j-1$, hence the expected information increment is

$$\delta I_{ij} \approx \rho_i A_{ij}.$$

Therefore the first-order evidence utility of allocating one extra call to contour i is

$$U_i^{(Z)} \propto \sum_{j>i} \frac{\rho_i A_{ij} g_j^2}{I_j^2}.$$

This score is large when contour i can reach boundaries that are both evidence-sensitive (g_j^2 large) and currently poorly learned (I_j small).

3. Objective 2: improve posterior resolution

Let n_j be the current number of effective posterior-support points in shell j . A simple resolution-loss proxy is

$$V_{\text{post}} \approx \sum_j \frac{m_j^2}{n_j + \kappa},$$

with a small stabilizer $\kappa > 0$. This says that shells with large posterior mass m_j need more support points to keep Monte Carlo error small.

A draw from contour i lands in shell j with probability P_{ij} , so one extra call at contour i has posterior utility

$$U_i^{(P)} \propto \sum_{j>i} \frac{P_{ij} m_j^2}{(n_j + \kappa)^2}.$$

This score is large when contour i feeds posterior mass into shells that matter to the posterior and are currently under-resolved.

If the target is a particular posterior expectation $\mathbb{E}[f \mid y]$, replace m_j^2 by $m_j^2 \sigma_j^2(f)$, where $\sigma_j^2(f)$ is the within-shell variance of f . Then the rule targets shells that are both posterior-important and variable for the chosen observable.

4. Choosing S contours at a time

Let $b_i \in \mathbb{N}$ be the number of the next S calls allocated to contour i , with $\sum_i b_i = S$. A natural batch objective is

$$\min_{\{b_i\}} \left[\lambda_Z \sum_j \frac{g_j^2}{I_j + \sum_i b_i \rho_i A_{ij}} + \lambda_P \sum_j \frac{m_j^2}{n_j + \kappa + \sum_i b_i P_{ij}} \right],$$

where $\lambda_Z, \lambda_P \geq 0$ set the relative importance of evidence precision and posterior resolution. The greedy marginal gain for adding one more call at contour i is then

$$\text{score}_i(\mathbf{b}) = \frac{1}{c_i} \left[\lambda_Z \sum_j \frac{\rho_i A_{ij} g_j^2}{(I_j + \sum_\ell b_\ell \rho_\ell A_{\ell j})^2} + \lambda_P \sum_j \frac{P_{ij} m_j^2}{(n_j + \kappa + \sum_\ell b_\ell P_{\ell j})^2} \right].$$

A simple allocation rule is:

- initialize $b_i = 0$ for all contours;
- repeatedly choose the contour with the largest current $\text{score}_i(\mathbf{b})$;
- increment its b_i by one and update the denominators;
- stop when $\sum_i b_i = S$.

The denominator updates create diminishing returns automatically, so the batch spreads across contours whenever one region becomes saturated.

5. Practical interpretation

The two objectives push to different contour regions:

- **Evidence only** ($\lambda_P = 0$): favor contours that sit high enough to inform many downstream boundaries with large evidence sensitivity.
- **Posterior only** ($\lambda_Z = 0$): favor contours whose descendants fall into high-posterior, currently under-sampled shells, including minor modes if their $m_j^2/(n_j + \kappa)^2$ score is large.
- **Mixed objective**: the greedy batch approximately equalizes marginal gain per unit cost across selected contours.

Finally, if posterior resolution is defined using only the original classic dead points, then extra contour calls do not directly increase that ESS. They only improve evidence and the resolution of the expanded sample tree. Thus it is useful to track two distinct goals: a classic-point ESS target and an expanded-tree utility target based on the objectives above.